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-- PART 3: DRIVER LOCATION HISTORY TABLE WITH PARTITIONING
CREATE TABLE IF NOT EXISTS driver_location_history (
  location_id BIGINT NOT NULL AUTO_INCREMENT,
  driver_id BIGINT NOT NULL,
  latitude DECIMAL(10,8) NOT NULL,
  longitude DECIMAL(11,8) NOT NULL,
  heading_degrees SMALLINT DEFAULT NULL,
  speed_kmh DECIMAL(5,2) DEFAULT NULL,
  accuracy_meters DECIMAL(8,2) DEFAULT NULL,
  recorded_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  ride_id BIGINT DEFAULT NULL,
  record_year SMALLINT NOT NULL,
  record_quarter TINYINT NOT NULL,
  PRIMARY KEY (location_id, record_year, record_quarter),
  KEY idx_dlh_driver_time (driver_id, recorded_at),
  KEY idx_dlh_coords (latitude, longitude),
  KEY idx_dlh_ride (ride_id)
) ENGINE=InnoDB
PARTITION BY RANGE COLUMNS(record_year, record_quarter) (
  PARTITION p2024_q1 VALUES LESS THAN (2024, 2),
  PARTITION p2024_q2 VALUES LESS THAN (2024, 3),
  PARTITION p2024_q3 VALUES LESS THAN (2024, 4),
  PARTITION p2024_q4 VALUES LESS THAN (2025, 1),
  PARTITION p2025_q1 VALUES LESS THAN (2025, 2),
  PARTITION p2025_q2 VALUES LESS THAN (2025, 3),
  PARTITION p2025_q3 VALUES LESS THAN (2025, 4),
  PARTITION p2025_q4 VALUES LESS THAN (2026, 1),
  PARTITION p2026_q1 VALUES LESS THAN (2026, 2),
  PARTITION p2026_q2 VALUES LESS THAN (2026, 3),
  PARTITION p2026_q3 VALUES LESS THAN (2026, 4),
  PARTITION p2026_q4 VALUES LESS THAN (2027, 1),
  PARTITION p_future VALUES LESS THAN (MAXVALUE, MAXVALUE)
);

CREATE TABLE IF NOT EXISTS ride_status_history (
  history_id BIGINT NOT NULL AUTO_INCREMENT,
  ride_id BIGINT NOT NULL,
  from_status
ENUM('requested','accepted','driver_en_route','in_progress','completed','cancelled','disputed')
  DEFAULT NULL,
  to_status
ENUM('requested','accepted','driver_en_route','in_progress','completed','cancelled','disputed') NOT
  NULL,
  changed_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  changed_by_user_id BIGINT DEFAULT NULL,
  changed_by_driver_id BIGINT DEFAULT NULL,
  notes TEXT DEFAULT NULL,
  latitude DECIMAL(10,8) DEFAULT NULL,
  longitude DECIMAL(11,8) DEFAULT NULL,
  PRIMARY KEY (history_id),
  KEY idx_rsh_ride (ride_id, changed_at)
) ENGINE=InnoDB;

CREATE TABLE IF NOT EXISTS fraud_flags (
  flag_id BIGINT NOT NULL AUTO_INCREMENT,
  entity_type VARCHAR(50) NOT NULL,
  entity_id BIGINT NOT NULL,

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flag_type VARCHAR(100) NOT NULL,  
severity ENUM('low','medium','high','critical') NOT NULL DEFAULT 'low',  
description TEXT NOT NULL,  
evidence JSON DEFAULT NULL,  
flagged_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
flagged_by VARCHAR(100) NOT NULL DEFAULT 'system',  
resolved_at DATETIME DEFAULT NULL,  
resolved_by VARCHAR(100) DEFAULT NULL,  
resolution_notes TEXT DEFAULT NULL,  
is_resolved TINYINT(1) NOT NULL DEFAULT 0,  
PRIMARY KEY (flag_id),  
KEY idx_ff_entity (entity_type, entity_id),  
KEY idx_ff_severity (severity, flagged_at)  
) ENGINE=InnoDB;
```